



Project:Advanced Drone Technology.

Project: Introduction to Guidance and Control Using Python Simulations

Objectives

The objective of this assignment is to help students:

1. Understand the fundamentals of Guidance and Control for autonomous moving systems.
2. Learn the basics of PID (Proportional–Integral–Derivative) controller tuning.
3. Observe the effect of environmental disturbances such as wind/current on system stability and tracking performance.
4. Develop intuition behind feedback-based control systems.
5. Understand trajectory/path-following concepts used in autonomous vehicles such as drones and boats.
6. Gain hands-on experience with Python-based simulations using Google Colab.

Environment Setup

Since this assignment is completely Python-based, we will use Google Colab to run the simulations. This avoids the need to install any software or dependencies on your system.

Follow the steps below to get started:

Step 1

Open the Google Colab notebook using the assignment link provided:

<https://colab.research.google.com/drive/1nUHfrkPvPUPY6uqAp6pwdO9kPMo3Inem?usp=sharing>

Step 2

Download the notebook file:



Files → Download →

Download .ipynb

Note:

If you already have a Python development environment such as Visual Studio Code, Spyder, Jupiter Notebook, or Anaconda installed on your system, you may also download the .py file and run it locally.

Step 3 Open a new browser tab and go to <https://colab.research.google.com/>

Step 4

Select: **Upload → Browse**

and upload the downloaded .ipynb notebook file.

Step 5

Once uploaded, the notebook will open in Google Colab and you can begin executing and modifying the code.

Procedure to Complete the Tasks

Task 1: Understanding PID Controller Design Using Drone Altitude Control

Objective

This task is designed to help students understand:

- The intuition behind feedback control
- PID controller implementation
- Gain tuning methodology
- The effect of wind disturbance on system stability and response



The drone in this simulation attempts to reach and maintain a desired altitude.

Instructions

1. Run the provided code cell.

Adjust the slider values for:

- Proportional Gain (P)
- Derivative Gain (D)
- Integral Gain (I)
- Wind Disturbance

Then click:

“Run Simulation and Plot”

to observe the response.

2. In the simulation:

- The **No Wind Zone** exists from **0 to 6 seconds**
- After 6 seconds, wind disturbance is introduced based on the wind value selected using the slider

Observe how the drone behaves before and after disturbance introduction.

3. While tuning the PID controller:

- First tune the **P gain**
- Then tune the **D gain**
- Finally tune the **I gain**



- Faster response
 - Minimum overshoot
 - Reduced oscillation
 - Better disturbance rejection
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4. Run the next notebook cell to view the drone animation corresponding to your tuned controller.

Observe:

- Stability
 - Oscillations
 - Recovery after disturbance
 - Overall smoothness of motion
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Task 2: Understanding Guidance and Path Tracking Using an Autonomous Boat

Objective

This task is designed to help students understand:

- Guidance algorithms
- Path-following behaviour
- Feedback correction
- Disturbance handling due to water currents

The autonomous boat attempts to follow a predefined trajectory.



Instructions

1. Run the notebook cells sequentially as instructed.

2. Use the sliders to tune the controller gains and environmental current values so that the boat follows the required path accurately.

The:

- **Blue dotted line** represents the desired trajectory/path
 - Boat trajectory shows the actual tracking performance
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3. Observe how different values affect:

- Tracking accuracy
 - Oscillation
 - Path deviation
 - Smoothness of motion
-

4. In the final notebook cell, manually update the following parameters using the values obtained from slider tuning:

- `kp`
- `kd`
- `current_x`
- `current_y`

Run the final animation cell to generate the accurate boat guidance animation.



Creative Bonus Task

Create a Python simulation in which a drone follows a **figure-eight trajectory**.

You may:

- Use waypoint tracking
- Use parametric equations
- Add disturbance handling
- Experiment with different controller gains

Bonus marks will be awarded for:

- Smooth trajectory tracking
 - Stable control
 - Good visualization/animation
 - Creativity in implementation
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Deliverables

Submit the following files/results:

1. Completed `.ipynb` or `.py` file
 2. PID Controller Tuning Result Plot
 3. Boat Guidance Plot:
 - With disturbance/current
 - Without disturbance/current
 4. (Optional) Figure-Eight Drone Simulation
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Important Note

For the **without disturbance/current** case:

```
current_x = 0  
current_y = 0
```

This represents ideal environmental conditions with no external current disturbance.